SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

MSc/PGD in Information Technology – 2026
Emerging Topics in Information Technology - IT6070
Assignment II

Title: AI Assisted Emerging Technology Software Solution

Assessment Weight: 30 marks

Assessment Type: Individual

Suggested Duration: Four weeks

Assignment Objective

Identify a small real-world problem faced by a named industry professional, then design, implement, demonstrate and evaluate a functional proof of concept using at least one emerging information technology.

Emerging technology must make a meaningful contribution to the solution. A conventional data-entry or CRUD application without a significant emerging-technology component is not sufficient.

This assignment assesses the following learning outcomes:

- **LO1:** Identify emerging technologies and concepts related to information technology.
- **LO2:** Explain the significance of emerging technologies.
- **LO3:** Propose and implement an integrated technological solution for a given requirement.
- **LO4:** Evaluate emerging technologies in relation to a given requirement.

Technology Areas

Students may select one or more areas, including:

- Artificial intelligence and machine learning
- Generative AI and large language models
- Intelligent web or mobile applications
- Internet of Things and edge computing
- Cloud and serverless computing
- Cybersecurity and privacy-enhancing technologies
- Blockchain and distributed applications
- Extended reality
- Intelligent databases and data analytics
- Emerging software engineering or automation technologies

Other suitable emerging technologies may be used with the lecturer's approval.

Assignment Task

Select a small real-world problem faced by an industry professional and develop a functional proof of concept that uses an emerging technology to address it.

Your solution must:

1. Name the selected professional, state their job title and industry, explain their role and working context, and describe the specific problem the solution will address. Obtain permission to identify the person; otherwise use a pseudonym and state this clearly. Explain how the problem was identified, such as a brief discussion or observation, and identify at least two requirements.
2. Explain why the selected emerging technology is suitable for this professional and problem; compare it with at least one feasible alternative.
3. Implement at least two connected functional features that support the professional's main workflow.
4. Integrate emerging technology into the main workflow.
5. Accept an input, perform meaningful processing and produce a visible output.
6. Include appropriate error handling or fallback behavior.
7. Test the solution with at least three suitable test cases, including an error or fallback case.
8. Evaluate the practical feasibility of adoption, including technical requirements, access to data or services, skills, privacy or security constraints, benefits, limitations and risks.
9. Estimate affordability for the selected professional or employer. Present an itemized cost estimate in a stated currency covering setup and recurring costs (such as devices, hosting, API usage, licences, maintenance and training); state assumptions, distinguish free-tier or simulated resources from realistic use, and compare costs with expected value or a lower-cost alternative.

10. Propose a realistic adoption or commercialization approach, including who would pay and the main condition for proceeding.

The implementation may use existing APIs, pretrained models, open-source libraries, cloud services or simulated data. Any simulation, limitation or externally provided component must be clearly identified.

Use of AI Tools

The appropriate use of AI tools is encouraged.

Students may use tools such as ChatGPT, GitHub Copilot, Gemini, Claude or other AI-supported development tools for:

- Generating and refining ideas
- Comparing technologies
- Designing the software architecture
- Producing or improving code
- Debugging and testing
- Generating test data
- Improving documentation

Students remain responsible for the correctness, security and quality of the submitted solution. AI-generated output must be reviewed, tested and modified where necessary.

The use of AI must be declared. Undeclared or misleading use of AI may be handled under the Institute's Academic Integrity Policy.

Required Deliverables

1. Software Implementation

Submit the complete source code as a ZIP file and accessible repository link.

The submission must include a README containing:

- Software requirements
- Installation and execution instructions
- Required API or configuration information
- Sample login details, if applicable
- Known limitations

Do not include passwords, private API keys or other confidential credentials.

2. Demonstration Video

Submit a narrated video with a maximum duration of **four minutes**.

The video must show:

- Student name and student ID
- The selected professional (name or declared pseudonym), role and specific problem
- The running software and its use in the professional's workflow

- The two main functional features
- How emerging technology is integrated
- One successful test and one error or fallback situation
- One key limitation and the estimated setup and recurring costs

Videos exceeding four minutes may be evaluated only up to the four-minute point.

3. Technical Evidence Document

Submit a maximum **four-page PDF** containing:

Page 1/2 – Solution Evidence

- Professional's name or declared pseudonym, role, industry, problem, evidence of need and two requirements
- Selected emerging technology, justification and one alternative
- Simple architecture diagram
- Two connected implemented features and their role in the workflow
- Three test cases and results
- Feasibility assessment, itemized setup and recurring cost estimates with assumptions, affordability judgment, benefits, limitations, risks and adoption approach

Page 3/4 – AI Use Declaration

Provide details of all important uses of AI:

AI tool	Purpose	Prompt or instruction	Output used	How the output was verified or corrected

Full AI conversation histories are not required.

Marking Rubric

Criterion	Excellent	Good	Satisfactory	Poor	Marks
1. Professional and Problem Definition **4 marks**	Names the professional (or declared pseudonym), role and industry; explains work context and credible evidence of the problem; gives two clear requirements. 4	Identifies professional, problem and two requirements, with minor gaps in context or evidence. 3	Provides a plausible professional and problem, but evidence or requirements are incomplete. 2	Professional or problem is unclear; requirements and evidence are weak or absent. 0–1	
2. Technology Selection **4 marks**	Accurately explains significance and fit of emerging technology; compares one feasible alternative against the professional's requirements. 4	Sound justification and comparison, with limited critical detail. 3	Relevant technology but weak explanation, fit or comparison. 2	Inappropriate technology or insufficient justification. 0–1	
3. Functional Implementation **8 marks**	Two connected features work reliably; valid input is processed to useful output in the professional's workflow. 7–8	Two features work with minor defects; core workflow completes. 5–6	One feature works or both works partially; workflow has significant defects. 3–4	Incomplete, unverifiable or interface-only solution. 0–2	
4. Integration and Software Quality **3 marks**	Emerging technology meaningfully drives the main workflow; architecture, usability, validation and fallback are appropriate. 3	Technology contributes to workflow; minor quality or error-handling gaps. 2	Integration is superficial, or software quality and fallback are weak. 1	No meaningful integration or usable workflow. 0	
5. Feasibility, Cost and Affordability **4 marks**	Evaluates technical, data, skills, time and privacy constraints; itemizes realistic setup and recurring costs with currency and assumptions; judges affordability against expected value or a lower-cost alternative; identifies an adoption path. 4	Covers main feasibility factors and setup/recurring costs; affordability judgment or assumptions lack some detail. 3	Basic feasibility discussion and rough costs; important cost categories, assumptions or affordability reasoning missing. 2	Feasibility or cost evidence largely absent or implausible. 0–1	
6. Testing and Critical Evaluation **3 marks**	Three test cases show input, expected and actual results, including an error case; critically discusses benefits, limitations, risks and improvement. 3	Three tests and useful evaluation, but evidence or critical depth is limited. 2	Limited tests or mostly descriptive evaluation. 1	Little or no reliable testing or evaluation. 0	
7. Responsible Use of AI **2 marks**	Declares all substantial AI uses with prompts, outputs used and evidence of verification or correction. 2	Declares AI uses but prompts, outputs or checking are incomplete. 1	AI use is missing, misleading or unsupported. 0	AI use is missing, misleading or unsupported. 0	
8. Demonstration and Submission **2 marks**	Video within three minutes shows the professional's problem, two features, technology use, success and error cases, and cost/limitation; code, README and PDF are accessible. 2	Submission is accessible but one or more required elements are unclear or incomplete. 1	Essential evidence, source or video is missing or inaccessible. 0	Essential evidence, source or video is missing or inaccessible. 0	
				TOTAL	